HUMEBOT · PROJECT REPORT 4

HumeBot: Open-Weight Models and Post-Training

Why the project studied open-weight language models, what it attempted, what it learned, and how the findings shape its direction

Mark Collier

Professor of Philosophy · University of Minnesota Morris

Audience	Philosophers, educators, AI researchers and technically interested readers
Version	1.0
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Project status	Research track; no open-weight model is qualified for teaching, and no post-training has been performed.

The HumeBot project website: https://mdcollier.github.io/humebot-site/

ABOUT THIS REPORT

Central question	What has HumeBot learned about choosing, running and improving language models for philosophical interpretation, and in particular about open-weight models and post-training?
Audience	Philosophers, educators, AI researchers and technically interested readers who know what generative language models are but not HumeBot's history. Report 1 describes the application, Report 3 the corpus, and Report 5 the experiments in full.
Evidence and scope	The project's model studies from August to November 2026: early evidence-use runs, the harness comparisons with Gemma 4 31B, a comparison with a second open-weight model, development and qualification work on the University's research computing, the post-training design and feasibility investigation, the reviewer-model pilots, and the frontier-model comparisons. Figures come from each study's closure or results record and were checked against the repository on 8 November 2026.
Status	No language model, open-weight or proprietary, is qualified or accepted for teaching. No post-training has been performed: no weights have been trained and no adapter exists. The open-weight line is a research track whose findings inform, but have not settled, the project's model strategy.

1. Why model choice matters for HumeBot

HumeBot is built so that the language model does not have the last word. Software constructs every displayed quotation from a controlled text, verifies every citation, keeps interpretation visibly separate from text, and withholds whatever fails its checks (Report 1). It would be a mistake to conclude that the choice of model therefore matters little. The harness guarantees the *provenance* of what a student reads; it does not supply the reading. Everything that makes an answer worth reading, which passages are selected, how an argument is reconstructed, whether a qualification is preserved or dropped, whether the model abstains when the evidence does not settle the question, is the model's work, and a weak model produces accountable but poor explanations, or none.

The project's requirements for a model follow from its three-stage pipeline. The model must **analyze** a student's question into the things an adequate answer has to establish; **select evidence** from the whole corpus, naming not only passages but the exact words within them that bear on each requirement; and **explain**, writing sentences each tied to specific words in the supplied evidence, within a strict structured format, and abstaining where the evidence falls short. Each stage sets a different demand. Evidence selection requires reading the entire pilot corpus, roughly 63,000 tokens of prompt, in one call, so the model needs a large context window and the ability to find relevant passages without an index. Explanation requires fidelity to the discipline's distinctions: scope, modality, what Hume asserts as against what he entertains or reports. And every stage requires reliable compliance with a machine-checked output contract, because a proposal the software cannot parse, or a quoted phrase that does not occur where the model says it does, produces no answer at all.

This divides the work cleanly. The harness enforces what can be enforced deterministically: that the text is Hume's, at the place cited, drawn from the evidence supplied, and tied to the sentence that cites it. The model is responsible for what no software can check: that the right passages were chosen, that the reconstruction is faithful, that nothing important was omitted. The project's model research is an inquiry into the second half of that division, and its central lesson is that the two halves fail differently.

2. Why open-weight models?

An **open-weight** model is one whose trained parameters, its *weights*, are published for download, so that anyone with suitable hardware can run it. Running a model to produce output is **inference**; the software that loads a model onto hardware and answers requests is a **model server**. A proprietary frontier model, by contrast, is reached only through its provider's service: its weights are not available, its exact version and settings are the provider's to change, and every request leaves the user's infrastructure. The project pursued open-weight models for five reasons, each with a practical limit.

Reproducibility. A study can name the exact model it used and run it again against identical weights, independent of a provider's update schedule. The project's standard for this is exacting: a served model is pinned by name, immutable revision and weight checksums, its numerical representation, the inference engine and its content-affecting settings, the tokenizer and chat template, and the served context limit. The limit, which the project states plainly, is that identical weights do not guarantee identical outputs; serving software, hardware and settings also matter, and repeated runs of the same configuration still varied somewhat.

Experimental control. With the model on hardware the project controls, one part of the system can be changed while everything else is held fixed, and everything the model is given can be logged. The development runs on the University's computing were made against a server reachable only from the same job, with no external model service in the loop.

Access to the model itself. Only an open-weight model can be post-trained by the project (section 4). The project's rulings make a practical path to future post-training a *requirement* on any model family it would qualify for teaching: a canonical trainable checkpoint, a licence permitting training and deployment of the result for University use, maintained support for parameter-efficient training, and an approved environment to train in.

Possible institutional hosting. An open-weight model could in principle run inside University infrastructure, keeping students' questions and the corpus off external systems. For most of the project's history this was the preferred classroom design, preferred but never institutionally approved; whether University infrastructure could provide a persistent, secure and responsive service for a course has not been established (section 5).

Cost and independence. Research inference on institutional hardware carries no per-token charge and depends on no commercial agreement, at the cost of the engineering and queueing that research computing involves.

The model at the centre of this work is **Gemma 4 31B** (`google/gemma-4-31B-it`), an open-weight instruction-tuned model of about 31 billion parameters released by Google under a permissive licence as declared on its model card; the project records that the terms of any model eventually deployed must still be reviewed by the University as part of that decision. From August to September 2026 the project used a hosted research inference service at a national laboratory, which was valuable for development but did not

expose the model's exact revision or serving configuration; the project limited it to development. From October 2026 the project ran the model itself at the University of Minnesota's Minnesota Supercomputing Institute (MSI): the canonical full-precision checkpoint, unquantized, on two NVIDIA H100 GPUs under an open-source inference engine, with a served context limit of 72,000 tokens and every element of the deployment's identity recorded. The practical limit was that of any research cluster: GPU time is scheduled in jobs rather than available continuously, so the arrangement suits experiments and not, as it stands, a classroom service.

Table 1. The models the project has exercised, where, in what role, and what the record shows. Judgments were the faculty lead's unless an AI reviewer is named in the text.

Model	Where	Role	What the record shows
Gemma 4 31B (open weights)	National-laboratory research service; then MSI	Working model throughout; frozen qualification candidate	Reads the whole corpus and follows the contracts; quotation fidelity is its principal weakness, contained by the harness; strongest open-weight configuration tested; not qualified
Llama 3.1 8B, 70B, 405B (open weights)	Research service	Early evidence-use runs	Mixed results; the 70B model's context window could not hold the corpus; not pursued further
gpt-oss-20b and gpt-oss-120b (open weights)	Research service	Early runs; 120B as the larger-model comparison	120B interpreted richly when prompted directly but misquoted heavily; delivered half the questions inside the harness; not pursued further
Qwen3.5-397B-A17B (open weights, 4-bit)	MSI	Writer only, in a capacity diagnostic	Repaired evidence-use omissions, not scope errors; no successor built
Nemotron-3-Ultra and gpt-oss-120b (open weights)	Research service	Candidate reviewers of HumeBot answers	One promising in a screen but unspecific in a pilot; no reviewer model adopted
Claude Opus 5.5 and Sonnet 5.5 (proprietary)	Provider API	Frontier comparisons; research configurations; recorded demonstration answers	Exact citations by construction; no fidelity gain from the harness in a small sample (Report 5)

3. The development experience

3.1 Can models use supplied evidence?

The project's first question was not which model was best but whether open-weight models could do the task at all. In two early runs, models were given a question and the right passages, chosen in advance, and asked to answer from them under one frozen harness. A smoke run across six models established that the instrument worked; a larger study across five models and 24 cases, adjudicated case by case by the faculty lead, passed 105 of 120 model-case cells, with Gemma and the larger gpt-oss model strongest at 23 of 24. The project claimed no winner: evidence was supplied, not found, and the cases were few. But the run showed that several open-weight models could use controlled evidence responsibly, and a capability probe settled a practical point: Gemma could take the whole 216-unit corpus in one prompt, where the 70-billion-parameter Llama model's context window could not. Gemma became the working model.

3.2 What the harness comparisons showed about the model

The project's principal studies of Gemma, a 24-question comparison of the harness against direct prompting and a 60-question held-out study, belong to Report 5; what matters here is what they revealed about the model. Prompted directly with the whole *Enquiry* and asked to quote and cite exactly, Gemma produced 12 faulty citations in 51, across 11 of 24 questions, none detectable from the response.

Quotation fidelity is a model-level weakness. Inside the harness the same weakness appeared as fail-closed non-answers rather than faulty quotations: 22 of 24 and 50 of 60 questions were delivered, the rest withheld because the model's misquotation was caught rather than shipped. The model followed the three-stage structured contracts on every question, and the evidence-selection stage, reading the whole corpus with no index, found evidence for 61 of 63 requirements; its one identified weakness, a requirement about the relation between two passages left unresolved when both had been selected, is a weakness of reading, not retrieval. The project's recorded conclusion was that Gemma is "not excluded as an underlying HumeBot model" and that the study "is not sufficient to approve it for instructional deployment."

3.3 A second open-weight model

Did the result reflect Gemma in particular? On the same 24 frozen questions and protocol the project ran `gpt-oss-120b`, a larger open-weight model. Judged blind by two AI reviewers, it had the strongest interpretive profile of any configuration when prompted directly, and it also overclaimed more; 83 percent of its declared citations were faulty, against Gemma's 23.5 percent. Inside the harness it delivered 12 of 24 answers to Gemma's 22: the harness contained a much less reliable model, but half the questions went unanswered. Gemma remained the strongest open-weight configuration tested, which did not qualify it, and a further step up in model size was deliberately not recommended as the next move. The lesson generalized: under a strict evidence contract, a model's reliability in quoting and following structure matters as much as the quality of its prose, because the harness can only withhold what it cannot verify, not repair it.

3.4 Development on MSI: where the open-weight system fails

In October 2026, with the candidate frozen for qualification, the project opened a development phase on MSI against a 36-question development set kept strictly apart from the sealed qualification material. The system delivered a verified answer for 35 of the 36 questions. "Delivery is not adequacy," the baseline report says, and the grading bore it out: of the 35 answers, 16 had no material problem, 15 had a material problem, and 4 contained a serious scholarly error. The grades were given by a single grader (the AI development assistant that built the harness), not blind to condition and with no second review; the project treats them as development evidence only, and the faculty lead later adjudicated the disputed cases.

Three diagnostics followed, each changing one layer while holding the rest fixed (Table 2). The first asked what the evidence selector should be allowed to see. In the frozen candidate the selector receives the analyzer's structured requirements but not the student's question; showing it the question as well did not change how much of the essential evidence it found, but it changed *which* passages it chose, and in the adjudicated grades it removed three of the four serious errors and raised adequate answers from 16 to 21 of 35. The project built this into a development successor, explicitly not a qualification candidate, since any change to a frozen candidate is a new candidate requiring its own freeze and fresh qualification. The second diagnostic appended general guidance to the writer's prompt; it did not touch the mechanisms behind the serious errors.

The third diagnostic examined whether the remaining errors reflected limitations in model capacity. Keeping everything fixed except the writer, the project served a far larger open-weight model, Qwen3.5-397B (a mixture-of-experts model in 4-bit quantization, on four H100s), and let it write from the same frozen evidence. **Greater capacity improved evidence use without eliminating the errors that matter most.** Adequate answers rose from 21 to 25 of 35 and material problems fell from 13 to 9, but the remaining serious error stayed, and the record's conclusion is exact: capacity "repaired evidence-use omissions, not the scope errors around conditional or interim claims." A bigger writer used more of the evidence it was given; it did not read Hume's qualifications more faithfully. No successor was built from it.

Table 2. The MSI diagnostics on the 35-question development set, as adjudicated: answers judged adequate, with a material problem, or with a serious scholarly error.

Configuration	Adequate	Material	Serious
Frozen candidate (Gemma 4 31B)	16	15	4
Selector shown the student's question	21	13	1
Larger writer (Qwen3.5-397B) on the same evidence	25	9	1

Taken together, these diagnostics located the open-weight system's failures with some precision. Delivery and citation were solved by the architecture. Omissions of available evidence responded to model capacity and to what the selector was shown. The serious errors, overstating the scope or finality of a claim Hume made conditionally or provisionally, responded to neither, and they are exactly the distinctions the discipline cares most about (Report 2).

3.5 The qualification attempt

The project had designed, in advance, a formal qualification for the frozen open-weight configuration: 170 newly written questions authored in isolation, frozen thresholds for delivery, false abstention, serious error and omission, and a frozen deployment identity on MSI. In October 2026 the run was executed in full. Its evidence gate returned *not eligible*, because one condition had invoked an exception under a rule that had never been adopted, and the faculty lead closed the attempt: no scholarly review took place, no pass or fail was established, and the questions, having been seen, cannot be used again for a clean qualification. A redesigned attempt exists only as planning and would need a fresh isolated set of about the same size, whose supply is a recorded risk. What the attempt did establish is that the infrastructure of qualification, freezes, custody, gates and review instruments, exists and worked as designed, including by refusing to proceed (Report 5).

3.6 Open-weight models as reviewers

Open-weight models were also tried as scholarly *reviewers* of HumeBot's answers, to make systematic review of many answers feasible. A screen in October 2026 tested two candidates against planted faults: one was judged unsuccessful, and Nemotron-3-Ultra promising, detecting most planted reversals and all planted omissions with few false alarms. A diagnostic pilot of that model on 236 completed calls then found it highly sensitive, detecting every planted reversal and overclaim and exposing a weakness in one of the project's own prior adjudications, but unspecific, since none of six clean answers passed both of its checks. The pilot was small and not independently blinded. No reviewer model was adopted, and the question of whether a development-time AI critic is worth building remains open.

4. Post-training: motivations, investigations and lessons

4.1 What post-training is, and what HumeBot would use it for

Post-training is any further training of a model after its original pretraining: **fine-tuning** on examples of desired behaviour, for instance, or **preference learning**, which adjusts the model toward outputs that reviewers prefer. Full fine-tuning updates all of a model's weights; **LoRA** (low-rank adaptation) instead trains a small set of added matrices, an *adapter*, while leaving the base weights frozen, which makes training far cheaper and keeps the base model intact. The project's interest was narrow and governed from the start. It never intended to teach a model Hume: its rulings forbid training on Hume's text as a target, injecting Hume content into weights, and training a model to reproduce stored correct answers. A successor model may learn, from governance-approved expert feedback, *how to interpret, reason about and teach from controlled evidence*: substantive interpretive quality, general intellectual behaviour such as epistemic restraint and the separation of text from interpretation, and pedagogical and communicative behaviour. Hume's text may appear as grounding context in such training, never as the thing to be learned. Any post-trained model is a new candidate with its own freeze and a fresh held-out qualification; training confers no provenance authority, and every successor passes the same gates.

The project also sequenced the track: qualify a base model; characterize its weaknesses that are *repeatable* across runs; determine which are plausibly trainable; then run initial adapter or fine-tuning experiments on MSI; then consider preference methods. Named techniques are candidates, not choices; full-model training is not contemplated. A data-separation rule keeps development feedback, qualification evidence, instructor feedback, approved training datasets and future held-out sets apart, with explicit promotion as the only route between them. And the classroom system is intended to include an instructor interface for recording structured feedback on answers, as a feedback-collection capability without commitment to any training method; it is not built.

4.2 What was attempted

In October 2026, after the capacity diagnostic had shown that a larger writer repaired omissions but not scope errors, the faculty lead opened a design for **behaviour post-training** of Gemma 4 31B. The design's hypothesis was that parameter-efficient training on *non-Hume* grounded-synthesis tasks would transfer to HumeBot's synthesis stage. The behaviours targeted were evidence-use behaviours only: using all the evidence assigned, completing multi-part requirements, synthesizing across the selected evidence, and not declaring the evidence insufficient when it was not; the scope errors were to be watched as guards, never trained. The data were to be self-generated and verifier-filtered: tasks built from two public multi-hop question-answering datasets unrelated to Hume (MuSiQue and 2WikiMultihopQA), answered by Gemma itself in HumeBot's packet format, and kept only when a production-grade verifier accepted them, with no teacher model. About 8,000 tasks were expected to yield about 3,000 accepted training examples. The method was to be LoRA with modest rank, the evaluation a paired statistical test on held-out non-Hume tasks and a pre-registered review by the faculty lead of transfer to the Hume development questions, with a stopping rule written in advance.

The project's proposed training sequence requires a qualified base model before any actual training. The two pilots run in October were inference-only data-feasibility checks, not training, and they ran before the qualification attempt had been executed and closed. Each generated answers to 300 tasks, using less than four tenths of a GPU-hour on two H100s, to measure whether enough acceptable training examples could be produced. They were a gate for the data, and the data failed it. Projected from the pilots, the main

generation would have produced about 500 accepted examples of the central synthesis behaviour against a required minimum of 800, and about 280 examples of warranted abstention against a required 400. The audit of the pilot outputs showed why. The two datasets' answers were adequate for checking whether a final answer was right, but in many cases their supporting sentences did not actually establish the relation the answer depended on: in one pilot, 14 of 50 "gold" reasoning steps rested on a sentence that did not establish it. Measured against HumeBot's standard, that a cited passage must establish what the sentence citing it claims, those labels were not trustworthy enough to filter a training set from, and the verifier was finding the mismatch. On 20 October 2026 the investigation was closed: **stopped before training, for a training-data construction and feasibility failure. No model weights were trained, no adapter exists, and the main generation never ran.**

4.3 What the evidence established, and why it matters

Post-training has not improved HumeBot; there is no trained artifact that could have. What the investigation established is bounded and useful. For the two datasets selected and the filtering procedure used, HumeBot's evidence standard proved stricter than the standard embodied in the datasets' own labels, so that a training set meeting it could not be manufactured by filtering them; whether other datasets, other filtering procedures or other sources of examples would fare better was not tested, and the investigation does not rule them out. The behaviours that most need improvement in the open-weight system, faithfulness to scope and qualification, were deliberately outside the design's reach, because no one knew how to construct trustworthy examples of them without a scholar's judgment on each case. And the governance did its work: the design stayed on behaviour rather than content, the stopping rule was honoured, and nothing was trained on a hunch. The finding points toward the route the project's rulings already anticipated as the most promising source of training signal for the behaviours that matter: expert feedback on real answers, collected through an instructor interface under University governance. It is also the slowest route, and it is not the only one conceivable.

5. Open-weight and frontier models

The project's frontier-model studies are analysed in Report 5; their significance for model strategy is the subject here. In late October 2026 the unchanged HumeBot prompts were run on Claude Opus 5.5 through its provider's service, first on five questions chosen because Gemma had struggled with them, then on twenty previously used questions, with a blinded AI reviewer judging. The samples were small and the judgments those of an AI reader, not a philosopher; three things nonetheless followed for the project's thinking.

The architecture's guarantee transfers across models. Inside the harness the frontier model's citations were exact in every case checked, as Gemma's had been in the held-out study; the citation guarantee holds by construction. But the frontier model also quoted exactly when prompted directly in that sample, where Gemma had not. The harness's most clearly demonstrated contribution, catching and withholding misquotation, matters most for models that misquote; with a model that does not, its value lies in what it makes inspectable and accountable rather than in faults caught (Report 2).

The open-weight model's interpretive weaknesses were its own. On the five hard questions every usable frontier answer reached the top tier while none of Gemma's did; on the twenty, the harness did not improve the frontier model's fidelity over direct prompting, and the reviewer more often preferred a direct

form of explanation on pedagogical grounds, with answer length confounded. The gap between the two models in interpretive quality was large, and the harness neither caused nor closed it. Model capability, not architecture, set the ceiling on what students would read.

The trade-offs became concrete. The frontier configuration sent the question and the corpus to an external provider under that provider's terms, with the model's version and settings outside the project's control, and carried a per-answer charge; the open-weight configuration kept everything on University hardware with the version pinned to the checksum and no per-token cost, but delivered weaker interpretation and no persistent service. Because the frontier runs also used settings different from the frozen contract's, the project records every such result as a *frontier experimental configuration*, not a run of the candidate.

From these findings a proposal emerged: to continue open-weight research on the University's computing, where models can be held fixed, inspected and eventually post-trained, while considering frontier models for teaching, where interpretive quality was highest among the configurations the project evaluated. The project records this as a proposal, not an adopted division of labour and not a model choice; it sits in tension with the earlier plan for an open-weight classroom model and with the rule that a teaching model's family must leave a path to post-training, and the faculty lead has not ruled. The current deployment-planning assumption for a first course is an external inference API approved by the University (Report 1), with a University-hosted open-weight service as an alternative under consideration. Neither kind of model is treated as intrinsically preferable: the open-weight path offers reproducibility, control, institutional hosting and the only route to post-training, at a present cost in interpretive quality and operational readiness; the frontier path offers the strongest interpretation among the configurations the project has evaluated, at a cost in control, reproducibility, data handling and dependence on a provider. Which cost a course can bear is a decision the project has placed before the University rather than made (Report 6).

6. What the project has learned

What depends on the model. Whether the right evidence is selected, especially for requirements that relate two passages; whether the explanation uses all the evidence it is given; whether Hume's qualifications, conditionals and provisional claims survive into the explanation; and whether an answer is delivered at all under a strict contract. The open-weight system's serious errors cluster on the third, and neither a larger writer, nor a better-informed selector, nor general prompt guidance removed them. Nothing in the record suggests that this is a permanent property of open-weight models; it is a property of the configurations tested.

What the architecture guarantees independently of the model. That every displayed quotation is Hume's, at the place cited, drawn from the evidence supplied and tied to the sentence citing it; that text and interpretation are kept apart; that what fails verification is withheld; and that an instructor can see what was supplied, selected and displayed. These held for a 31-billion-parameter open-weight model and for a frontier model alike, and for a much less reliable model at the price of half its answers.

What remains uncertain. Whether any open-weight model can be qualified for teaching against the project's thresholds, since the one attempt did not reach review; whether behaviour post-training can be fed from any source short of expert judgment, since the one attempt to do so failed its own gate; whether the University's research computing could host a persistent service for a course; and which path, or what combination, the project should take.

Where further model research would be justified. Characterizing the frozen candidate's repeatable weaknesses, across runs rather than once, is the prerequisite the project's own sequence sets for any training. The selector change that removed most of the development set's serious errors is a candidate for a successor, through the full freeze-and-qualify discipline. A fresh, isolated qualification set is the gate on any claim that an open-weight configuration is adequate for students. And structured faculty feedback on real answers, under University governance, is the most promising source of training signal for the behaviours that matter.

Conclusion

HumeBot's open-weight research was pursued for sound reasons, reproducibility, control, institutional hosting and the possibility of post-training, and it produced a precise result rather than a product. An open-weight model of moderate size can read the whole pilot corpus, follow a strict evidence contract and, inside the harness, cite Hume without fault; its failures lie in interpretation, most seriously in the preservation of Hume's qualifications, and a larger writer repairs omissions but not those. Post-training toward better evidence behaviour could not be fed from the public datasets tried, and nothing has been trained. The central lesson is the one the whole collection turns on: the verification guarantees transfer across models, interpretive performance depends on the model's capabilities, and no model, open-weight or proprietary, has yet been qualified for teaching.

Sources and document status

- Strategic directions §§14.4–14.9, 14.12 and 14.13 (the open-weight finding; the larger-model comparison; the Spring 2027 track and candidate infrastructure roles; post-training as a research track; the post-training rulings PT-1 to PT-4) and §15.4 (the research track); the constitutional document §2 and §4.
- Evidence-use runs: the oracle smoke run and the Stage A oracle study (closed 13 September 2026); the whole-corpus capability probes.
- Native harness comparison (September 2026), the held-out generalization study and the model comparison with gpt-oss-120b (23 September 2026), as summarized; see Report 5.
- Qualification deployment record and freeze (MSI, October 2026); the development baseline, selector-information and synthesis ablations, the development successor and the synthesizer capacity diagnostic (MSI, October 2026).
- Instructional Qualification 0.1 decision record (design, freezes, execution and closure, September–October 2026) and the IQ-0.2 design memorandum.
- Behaviour post-training design memo and the post-training feasibility investigation's closure record and pilot findings (20 October 2026); the model-family assessment for Gemma 4 31B.
- Reviewer research: the reviewer probe 0.1 and pilot 0.2 (October 2026) and the Stage 0 documentary study.
- Frontier pilot 0.1 and frontier comparison 0.2 (29 October 2026); research roadmap 0.1 (proposal); see Report 5.

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